

From Relevance to Reward: A Pipeline View of Addictive Design in Recommender Systems under the DSA

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Recent EU Commission proceedings against TikTok, Meta, Shein, and Temu have framed addictive design as a systemic risk under the Digital Services Act, yet the connection to recommender systems remains underexplored. This contribution proposes a four-stage analytical framework centered on relevance estimation, session sequencing, objective configuration, and interface and notifications, mapping where addictive effects emerge in the recsys pipeline and how they can be audited. Drawing on the 2025 DSA audit and systemic risk assessment reports of six VLOPs, we identify gaps in current transparency practices, including no disclosure of optimisation objectives and impact-relevant metrics, and derive recommendations for DSA enforcement.

Keywords: Recommender Systems, Addictive Design, Digital Services Act

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1 Introduction

Recently, the harms deriving from the compulsive use of social media and other online intermediaries have come under scrutiny in the European Union. In February 2026, the EU Commission issued the preliminary conclusions of an investigation into TikTok for its compliance with the Digital Services Act (DSA), according to which the company "did not adequately assess how [...] addictive features could harm the physical and mental wellbeing of its users, including minors and vulnerable adults. The main addictive features identified by the Commission include "infinite scroll, autoplay, push notifications, and its highly personalised recommender system" [4]. This combination, according to the Commission, generates a vicious circle of hook and reward, leading the user to compulsive use, impairing self-determination and bypassing their decision-making.

Notably, in the preliminary conclusions the Commission highlighted that:

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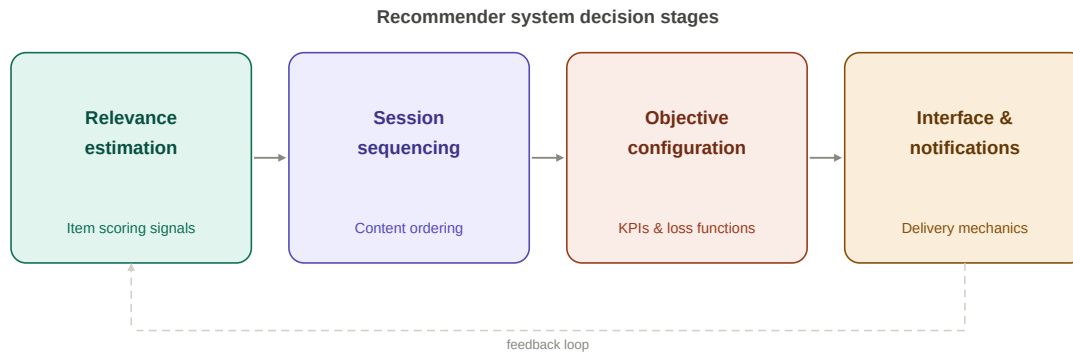


Fig. 1. The four recommender system decision stages through which additive design effects can be introduced or mitigated, and their feedback loop.

- (i) TikTok disregarded important indicators of addiction, including the frequency with which users open the app and the duration of night-time use by minors;
- (ii) the screen-time management tools and parental control tools are not effective mitigating measures, as they either don't create enough friction or require time and skills that parents and average users might not have.

As part of the proceeding, TikTok can file a reply to the Commission's preliminary findings. If, however, at the end of the proceeding the findings will be confirmed, TikTok could be obliged to change the design of its platform, by eliminating infinite scroll and adapting its recommender system. The Commission's approach, while correctly framing the role of both design and recommender systems in generating online harms, does not specify which parameters of addiction should be considered in performing the systemic risk assessment, which mitigating measures can take the place of the existing ones and, in particular, how can the recommender systems be adjusted to avoid addiction.

The connection between addictive design and recommender systems, while not explicitly mentioned in the DSA, is addressed by a 2023 EU Parliament resolution calling for an "assessment by the Commission of the addictive and mental-health effects of interaction-based recommender systems, in particular hyper-personalised systems, that keep users on the platform as long as possible rather than serve them information in a more neutral manner" [5]. The Parliament asked the Commission to "clarify the risk assessment and mitigation of [...] potential harms to health caused by the addictive design of recommender systems" from the perspective of the enforcement of the DSA systemic risk provisions (art. 34 and 35), proposing user control for "meaningful personalization" and interoperable recommender systems [5].

Having opened proceedings against TikTok, Meta, Temu and Shein for addictive design, the Commission has begun to consider this issue a priority for the DSA enforcement. However, the connection between "hyper-personalized" recommenders and addictive design is still underexplored. Grounded in the review of publicly

available evidence on platforms' compliance strategies, including the 2025 DSA audit and systemic risk assessment reports, this contribution aims to map how the concept of addictive design can be operationalized with regard to recommender systems. The analysis builds upon the DSA requirements on the transparency and controllability of recommender systems (art. 27 and 38), dark patterns (art. 25), protection of minors (art. 28 and related guidelines), systemic risk identification (art. 34) and mitigation (art. 35).

2 Analytical Framework

We identify four decision stages at which a platform based on recommender systems can produce or mitigate addictive design effects. These stages correspond to distinct design choices of a modern recsys pipeline. Together, they constitute the key leverage points at which platform design choices translate into user behavioral outcomes.

Relevance estimation concerns how a system scores candidate items for a given user; it can be operationalized in different ways [2]. This stage is foundational because the choice of input signals, including engagement-based signals such as watch time, re-opens, and repeat exposure, directly shapes which psychological triggers are amplified. Research on variable-ratio reinforcement schedules [6] and on the neurological mechanisms of social reward [7] shows that relevance models optimising for short-term engagement implicitly exploit dopaminergic feedback loops. Auditing this stage requires disclosure of which signals are used and their relative weights, a disclosure currently mandated but rarely provided in informative terms under DSA Article 27.

Session sequencing refers to the ordering and aggregation of content across a session, including the interleaving of different content verticals (video, image, story) and the triggering of autoplay or infinite scroll mechanics. Even a relevance model with modest individual-item scores can produce compulsive consumption when items are arranged to eliminate natural stopping points [3]. This stage is where architectural features like infinite scroll and autoplay have a strong impact: they are sequencing decisions, not only interface ones. Auditing session sequencing requires data on session-length distributions and the presence or absence of friction in content transitions.

Objective configuration concerns the global optimisation targets of the recommender, i.e. the KPIs and loss functions the system is trained or tuned to maximise. This is arguably the most consequential decision stage, because it determines what behaviour the system is ultimately rewarded for producing. Platforms optimising for raw engagement metrics such as daily active users, session length, or return frequency have structural incentives to produce compulsive use, irrespective of any downstream mitigation. Obermeyer et al. [9] and the broader algorithmic fairness literature have documented how optimisation objectives propagate through model training in ways that are difficult to correct post hoc. Critically, this stage requires no new data: the platform already knows what it optimises. Auditing requires only that this information be disclosed and assessed against wellbeing criteria.

Interface and notifications encompass the delivery mechanics through which algorithmic outputs reach users: visual presentation, notification triggers, timing, and the design of any interstitial friction. This is the stage most visible to regulators and users, and the focus of most existing DSA proceedings. However, interface-level mitigations might remain structurally limited if upstream optimisation objectives remain unchanged. Auditing this stage should assess not only whether tools exist, but whether they meaningfully alter usage behaviour – a measurement largely absent from current platform disclosures.

The four stages interact in a feedback loop: interface signals (e.g., notification open rates, session restart patterns) are fed back into relevance estimation, closing the cycle. This architecture means that addictive design is not a

property of any single component but an emergent property of the pipeline as a whole, which is precisely why a stage-by-stage auditing framework is needed. The framework complements and extends prior work on the ethical challenges of recommender systems [1, 8] by grounding abstract normative concerns in the concrete technical decisions that DSA audits must address.

3 Methodology and Expected Outcomes

Our empirical basis consists of DSA-mandated documentation published in 2025 for six VLOPs currently undergoing DSA investigation or proceedings regarding addictive design and/or recommender systems: TikTok, Meta (Facebook and Instagram), YouTube, Shein, and Temu. For each platform, we analyse the annual audit report and the platform’s own systemic risk assessment, yielding a corpus of twelve documents. The analysis proceeds in two stages. In the first stage, we perform passage retrieval to identify all text segments relevant to recommender system design and addictive design risks. In the second stage, each retrieved passage is annotated along two orthogonal dimensions: (i) the decision stage it addresses (relevance estimation, session sequencing, objective configuration, or interface and notifications) and (ii) the function it serves within the risk governance cycle: risk identification, risk assessment, mitigation, or measurement. This annotation scheme allows systematic cross-platform comparison while remaining grounded in the technical framework described in Section 3.

The expected contributions are threefold. First, we provide a structured catalogue of how platforms and their independent auditors currently frame and respond to addictive design, enabling the first cross-platform comparison anchored in recsys pipeline stages. Second, we identify key gaps in the transparency and accountability mechanisms currently supported by the DSA, as reflected in how VLOPs operationalise their obligations in practice — most notably, the absence of any disclosure of recommender system optimisation objectives in relation to user wellbeing, and the near-total lack of effectiveness evidence for proposed mitigations. Third, we derive targeted recommendations for DSA enforcement, clarifying how the related provisions should be interpreted to mandate disclosure at each decision stage.

References

- [1] Himan Abdollahpour, Gediminas Adomavicius, Robin Burke, Ido Guy, Dietmar Jannach, Toshihiro Kamishima, Jan Krasnodebski, and Luiz Pizzato. 2020. *Multistakeholder Recommendation: Survey and Research Directions*. Vol. 30. 127–158 pages. doi:10.1007/s11257-019-09256-1
- [2] Aparna Balagopalan, Abigail Z Jacobs, and Asia J Biega. 2023. The role of relevance in fair ranking. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2650–2660.
- [3] Amanda Baughan, Mingrui Ray Zhang, Raveena Rao, Kai Lukoff, Anastasia Schaadhardt, Lisa D. Butler, and Alexis Hiniker. 2022. “I Don’t Even Remember What I Read”: How Design Influences Dissociation on Social Media. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI ’22)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3491102.3501899
- [4] European Commission. 2026. Commission preliminarily finds TikTok’s addictive design in breach of the Digital Services Act. <https://digital-strategy.ec.europa.eu/en/news/commission-preliminarily-finds-tiktoks-addictive-design-breach-digital-services-act>
- [5] European Parliament. 2023. Addictive design of online services and consumer protection in the EU single market - European Parliament resolution of 12 December 2023 on addictive design of online services and consumer protection in the EU single market (2023/2043(INI)). https://www.europarl.europa.eu/doceo/document/TA-9-2023-0459_EN.pdf
- [6] Charles B. Ferster and B. F. Skinner. 1957. *Schedules of Reinforcement*. Appleton-Century-Crofts, New York.
- [7] Dar Meshi, Diana I. Tamir, and Hauke R. Heekeren. 2015. The Emerging Neuroscience of Social Media. *Trends in Cognitive Sciences* 19, 12 (2015), 771–782. doi:10.1016/j.tics.2015.09.004

- [8] Silvia Milano, Mariarosaria Taddeo, and Luciano Floridi. 2020. Recommender Systems and Their Ethical Challenges. *AI & Society* 35 (2020), 957–967. doi:10.1007/s00146-020-00950-y
- [9] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* 366, 6464 (2019), 447–453.